

APPENDIX L1:

The impact of light on the circadian system can be measured using a metric called Equivalent Melanopic Lux. This metric was proposed by Lucas and others (Lucas et al., "Measuring and using light in the melanopsin age." Trends in Neuroscience, Jan 2014). The authors provided a toolbox which for a desired spectrum derives equivalent "α-opic" lux for each of the five photoreceptors in the eye (three cones, rods, and the ipRGCs).

EML is dependent on the light intensity as well as the spectral power density of light at the measurement point. Given a spectrum of light, each equivalent α-opic lux is related to each other by a constant called Melanopic Ratio (R).

To calculate the equivalent melanopic lux (EML), multiply the photopic lux (L) designed for or measured in a building by this constant (R): $EML = L \times R$.

Projects may use spreadsheets published by the [authors of the journal article](#) or [IWBI](#) for this calculation.